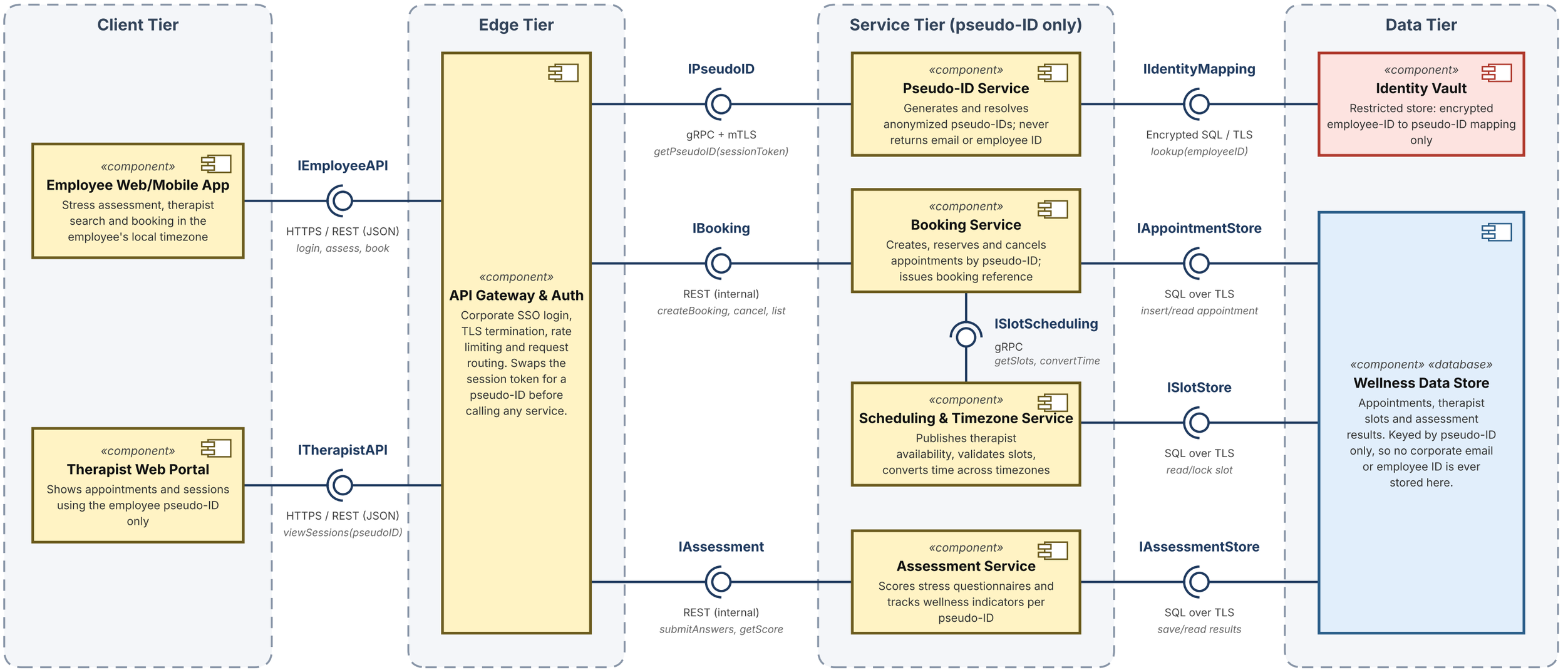
Corporate Mental Wellness & Counseling Platform — UML Component Diagram

Architectural style: Microservices | Lab 3 · Problem Statement #14



Legend

Provided interface (ball) — Required interface (socket) — Assembly connector (consumer socket wraps provider ball) — Restricted: holds identity

Privacy rule: only Pseudo-ID Service and Identity Vault ever touch corporate email / employee ID. Every other service and the therapist portal see the pseudo-ID only.

Architectural Justification

Problem Statement #14 – Corporate Mental Wellness & Counseling Platform

Name: Niranjan S SRN: PES1UG24AM177 Section: C

Architecture Selection

We chose Microservices Architecture for the Corporate Mental Wellness & Counseling Platform.

Architectural Choice

The platform is split into small, independently deployable services behind an API Gateway: a Pseudo-ID Service, a Booking Service, a Scheduling & Timezone Service and an Assessment Service. Each service talks to the others and to the Wellness Data Store only through the provided and required interfaces shown in the component diagram (for example IBooking, ISlotScheduling and IPseudoID). A Layered design would let every business-layer module reach identity data and could only be scaled as one block. A Client-Server design concentrates identity and clinical data on one central server, which is a single point of failure.

Two Reasons for This Choice

1. **Identity isolation enforces the core privacy rule.** The therapist must never see the employee's corporate email or employee ID. With microservices, identity data is confined to exactly two components, the Pseudo-ID Service and the Identity Vault. The Booking, Scheduling and Assessment services and the Therapist Portal only ever receive the pseudo-ID, so the rule is enforced by the structure of the system and not by developer discipline.
2. **Features have different workloads and change at different rates.** Slot lookup, timezone conversion and booking are busy at peak times across global offices, while stress assessments are used far less often. Separate services can be built, deployed, scaled and replaced independently. For example, the timezone logic can be upgraded behind ISlotScheduling without touching the Booking Service.

Security Advantage

Smaller blast radius through least privilege. Only the Pseudo-ID Service may call the Identity Vault, over mutually authenticated TLS (mTLS), and the API Gateway handles corporate SSO login and swaps the session token for a pseudo-ID before any request reaches a service. If the Booking Service, the Scheduling Service or the Wellness Data Store were compromised, the attacker would obtain pseudo-IDs only, which cannot be linked back to an employee without also breaching the separately protected vault.

Performance Benefit

Independent horizontal scaling. At peak booking times, additional instances of the Scheduling and Booking services can be started without scaling the rarely used Assessment Service, and the Scheduling Service can cache therapist availability per timezone to keep slot lookups fast. Fault isolation is a related benefit: if the Assessment Service goes down, employees can still book sessions.

Trade-off Acknowledged

Microservices add network latency and operational complexity. We limit this by keeping the number of services small (four) and coarse-grained, using efficient internal protocols (gRPC) for service-to-service calls, and routing all client traffic through one API Gateway.